Generative AI in the Business Setting

A practical guide to using AI to advance your understanding of business

October 2024

Lundquist
College of Business

Learning Objectives

What Students Should be Able to Do

Practical Application of AI Tools

Students will demonstrate proficiency in using AI tools and platforms to analyze data, develop models, and create AI-driven solutions to real-world problems including the development of an AI product.

Innovation and Problem-Solving with AI

Students will be capable of applying AI techniques to innovate and solve complex problems in various domains, recognizing the potential benefits and limitations of AI in the business setting.

Responsible, Ethical and Societal Implications

Students will critically assess the ethical considerations and societal impacts of AI technologies, understanding the importance of responsible AI development and deployment.

Collaboration and Communication in AI Projects

Students will effectively collaborate in multidisciplinary teams to design, implement, and present AI-driven projects, clearly communicating the benefits and challenges of AI to both technical and non-technical audiences.

Criteria for Earning the Microcredential

- *Completed 8 hours of training*
- *Participate in surveys about Generative AI uses*
- *Participate in ethical and responsible AI sessions*
- *Create team one-pager for new AI Product*

Agenda (Revised)

- What is Generative AI?
- Team Building
- Class Exercise
- AI in Business
- AI Product Development
- Ethics and Responsibility
- Contracting & Consulting
- Validation and Regulation
- Financial Analysis
- Resource Management
- Presentations

Day 1 - What is AI

4 hours

What is Generative AI?

Surveys on Mentimeter

Mini Lecture | Team Exercise | Discussion | Informational

30 min

Paper Clips

Use the following prompt to create 50 uses of paper clips. Acceptable tools, ChatGpt, Meta, Perplexity

“as a designer, I want you to come up with 50 uses of paper clips. They need to be creative and practical"

“Sure! Here’s the list:

1. Cable Organizer
2. Bookmark
3. Keyring Substitute
4. Zipper Pull Replacement
5. Reset Tool
6. Earring Back
7. Bag Sealer
8. Phone Stand
9. DIY Bracelet or Necklace Clasp
10. Paint Stirrer”

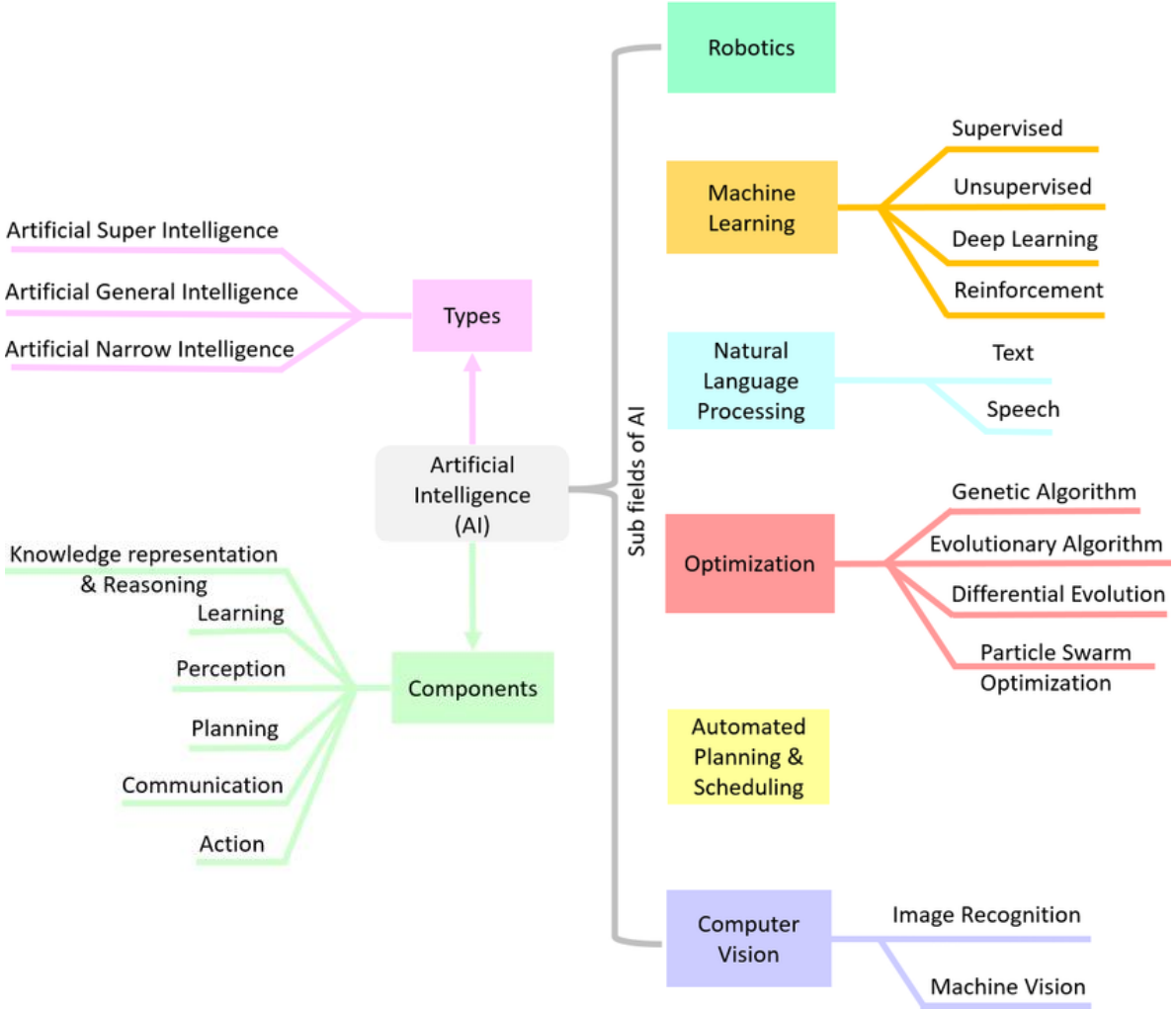
What is Generative AI

- AI consists of many components

AI is a compilation of tools and practices that make up different capabilities.

Generative AI is a set of components that learn, reason, and predict what you are thinking.

Some of these include RNNs, GANs, NLP, and machine learning



Teams

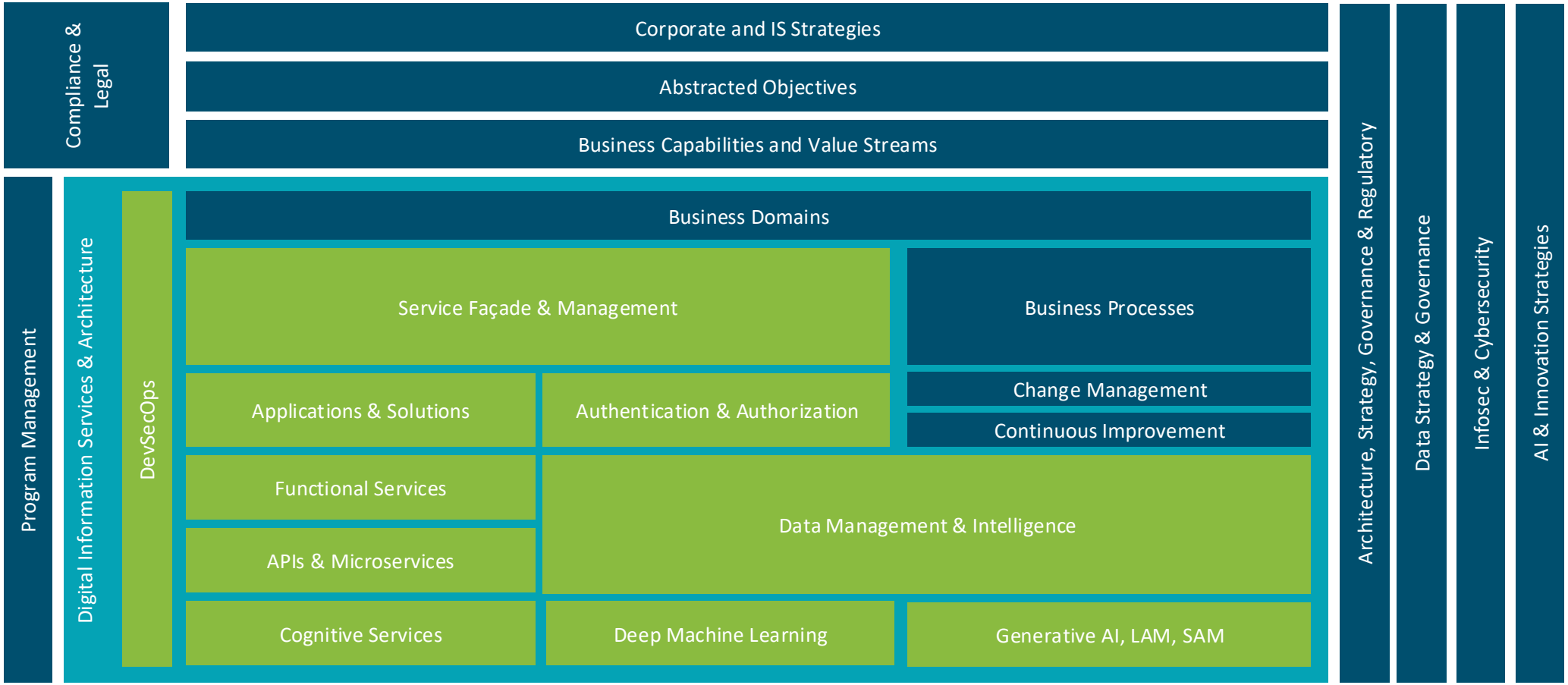
- Self-organize
- Assign a scribe
- Assign a lead
- All contribute
- All present

Guidelines

- Team uses AI to develop a product
- Can use any combination of tools
- Must be beneficial in a business setting
- Must show value
- Everyone must present

AI in Business Primer

Innovation Framework – AI Product Development – New App



Roles
C-level executives
SVP, VP, AVP
Sr. Directors, Architects
Product Owners, Business Analysts
Architects, Analysts, Developers
Data Architects, Engineers
Cybersecurity Infosec Engineers
AI Architects, Developers

10 Min Break

AI Product Development

AI Product Definition and Examples

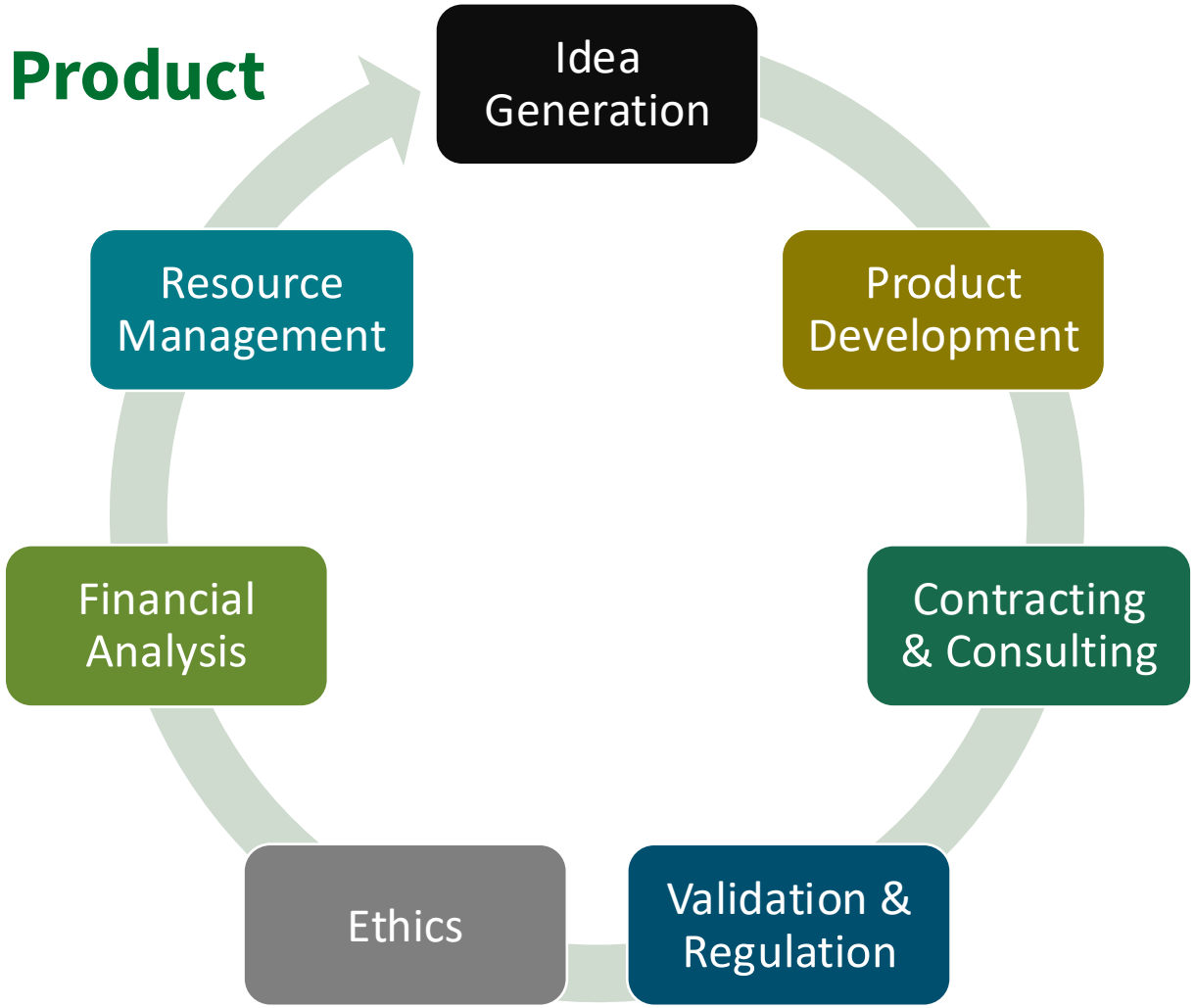
An AI product is a tool or app that uses smart technology to perform tasks that usually need human thinking, like understanding speech or recognizing images.

Which student is the AI?

Overview: Design an AI-powered Product

- Using Generative AI, create a new product and identify how each of the following areas will be impacted.

- Remember, you are the **producer**, Generative AI is your **assistant**.



Corporate and IS Strategies

Abstracted Objectives

Business Capabilities and Value Streams

Perspective – Business, Industry, Employee

Ideate, design, and create an AI Product

- Each team will decide on an **AI Product**
- To start the exercise, as a **business** team, each team needs to create 4 things, a business case, tagline, 30 second pitch, list of potential jobs for that product
- Teams are presented with prompts, use these prompts to help do this, also update them to make the experience unique
- After the initial artifacts, teams need to evaluate the **industry** in relation with their AI product
- Finally, from the viewpoint of the **employee**, the team will need to use AI to write a job description and a resume for that job description and expand on the skills an employee must possess

Team Selection – Pick a Product

Product Name	Industry	Description
BioSynth Insight	Life Sciences	Analyzes clinical trial data to identify trends, predict outcomes, and optimize drug development timelines.
DrillSmart AI	Oil & Gas	Predicts optimal drilling locations by analyzing geological and environmental data.
VentureVision AI	Venture Capital	Evaluates startups using market trends, pitch analysis, and financial projections.
BrandMuse AI	Brand Development	Generates slogans, logos, and branding concepts tailored to specific audiences.
MarketMosaic	Business Strategy	Combines competitor analysis, consumer behavior, and economic data for strategic planning.
CarbonTrack Pro	Sustainability	Measures carbon footprints, suggests reduction strategies, and tracks sustainability goals.

Compliance & Legal	Architecture, Strategy, Governance & Regulatory
	Data Strategy & Governance
	Infosec & Cybersecurity
	AI & Innovation Strategies

Generative AI in Business

Exercise: Each team creates an AI product

Human: Decide on a digital AI product

AI: Use prompts to create the following artifacts

- **Business case**
- **Tagline**
- **30 second pitch**
- **List of jobs needed to develop the idea**

Prompt: "**As a business analyst that is developing a digital product, <thing>, write a professional business case, tagline and 30 second pitch as though you were speaking to a group of investors and c-suite executives.**"

Generative AI in Industry

Exercise: Each team evaluates the trends in the industry

Human: Understand where your product fits in the industry

AI: Use prompts to identify the industries you might be able to use your product in

Prompt:

”**Now, acting as a marketing executive responsible for marketing new products to different industries, analyze my product and identify which industries it can be used in and in which ways, sort them in order of potential impact, create a table that describes these.**”

Generative AI for the Employee

Exercise: Using the list of jobs needed, each team member picks one and builds it out

Human: Using a job you would like to do or learn, use AI to help with a job description

AI: Use prompts to write a job description and resume outline based on the description

Prompt 1:

”My digital AI product requires <job>, write a job description of that <job> for <industry>. Then write a resume for the ideal candidate.”

Prompt 2:

“Now that you have the job description and resume, outline the top skills I need to learn to do this job and provide resources so I can learn this job. Also detail about how long it will take to learn them.”

Compliance & Legal	Architecture, Strategy, Governance & Regulatory
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Ethics and Responsibility

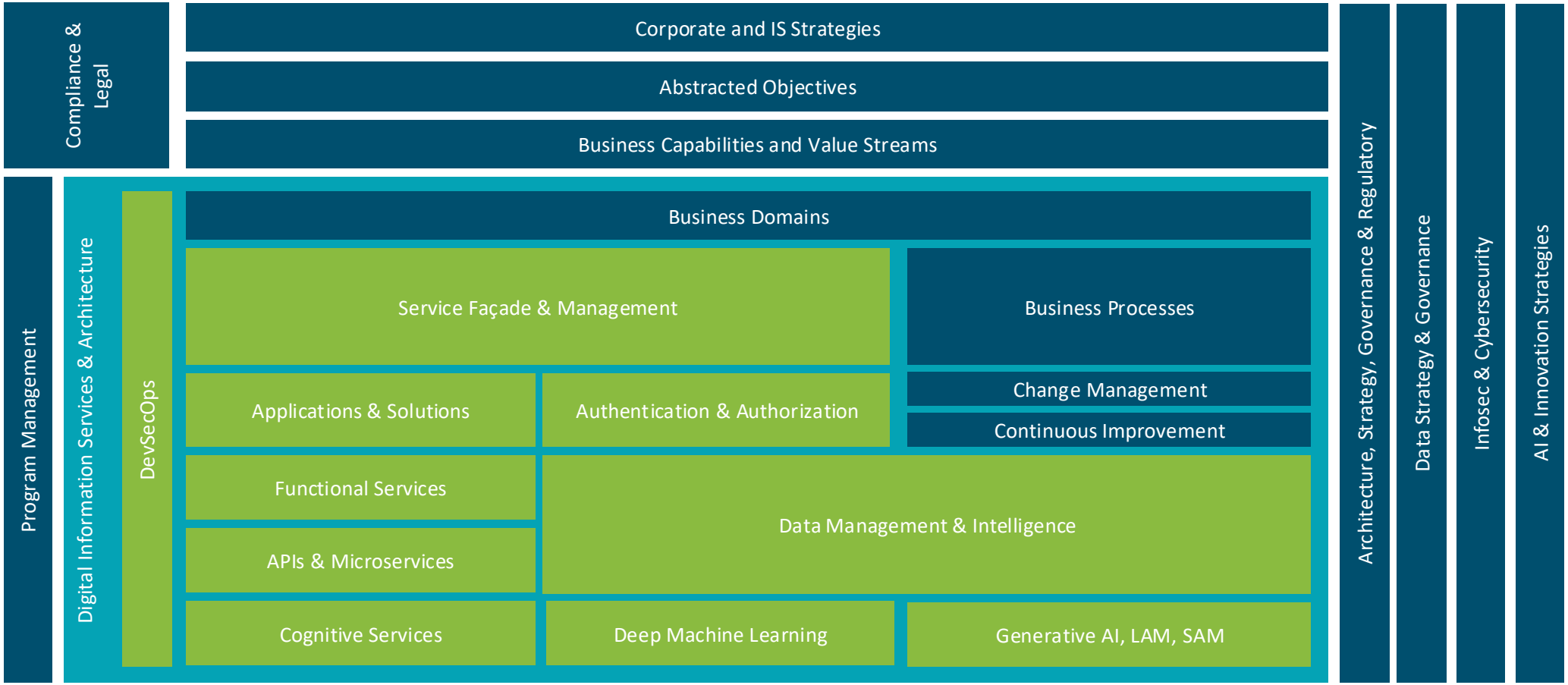
Mentimeter

End of Day 1

Day 2 – Product Development

4 hours

Innovation Framework – AI Product Development – New App



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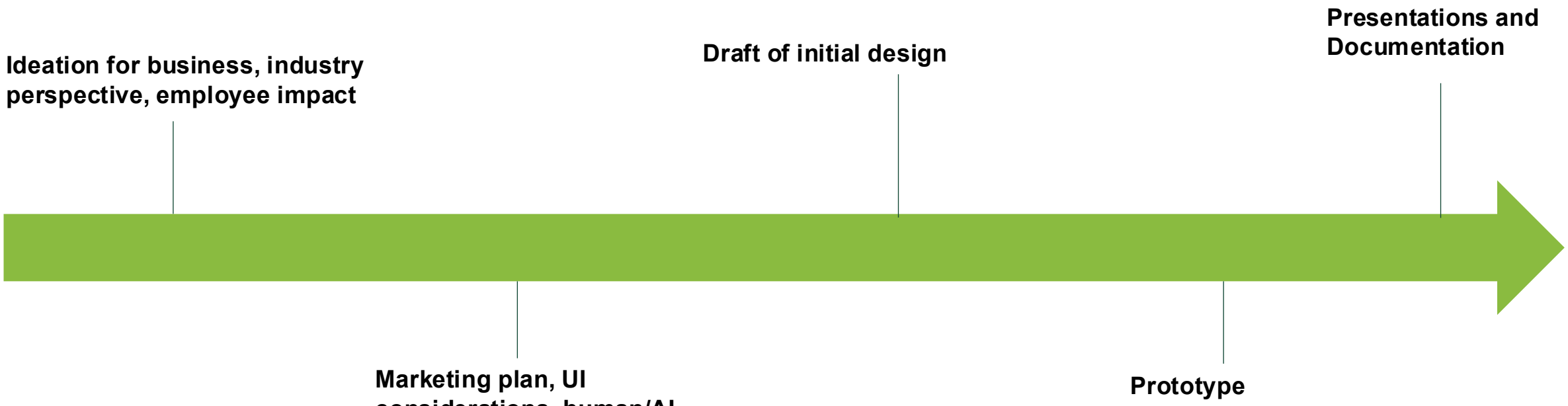
Mini Lecture | Team Exercise | Discussion | **Informational**

5 min

Product Development and Design – Document your AI Product

Create a presentation of your AI Product



Mini Lecture | Team Exercise | Discussion | Informational

5 min

Putting it All Together and Presentations

Use Generative AI to compile your business plan and present **one-pager** to the group

- Marketing plan
  - Terms and Conditions
  - Use of AI statement
  - Resource Model
  - Ethics and Responsibility statement
  - Logo and slogan
- Financial Analysis
  - Resource Management Plan
  - Development and Implementation Roadmap
  - Startup Costs
  - Initial Ask

Each team will compile this into a 1-pager and present it to the group. Choose a **different person** for each section.

Contracting & Consulting

Team will decide for their product if they will hire employees, and if so what types, hire a consulting firm, or hire contractors

- ❑ Employees, which ones
- ❑ Contractors
- ❑ Consultants

Consultant		Contractor
Provide organizations with guidance and support.	Role	Perform a specific task.
Has broad expertise of an industry.	Expertise	Has a specific skillset that doesn't require a complete understanding of an industry.
May have an ongoing relationship with clients, providing support over a long period of time.	Client Relationship	Responsible for completing a specific task or project within a set period of time.
Help organizations improve their performance or achieve their goals	End Goal	Deliver a specific product or outcome within a defined scope, timeline, and budget.

As referenced from https://blog.hubspot.com/sales/consultant-vs-contractor

HubSpot

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Validation & Regulation

Some of the AI products will require validation. Ask AI what validation or regulation requirements may be required with your AI Product

Prompt: "**I am designing and developing an AI product for <insert name and description>. Because this AI Product is in <industry>, please identify any regulation or validation requirements that may be necessary to take this AI Product to market.**"

Exercise

- ❑ Document list of regulations
- ❑ Use AI to summarize why needed
- ❑ Use AI to create a list of activities needed

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Ethics and Responsibility

Every AI Product should identify risks and ethical considerations.

Prompt: "**My AI Product <name of product> is focused on <describe what it is focused on> and has inherent risks and ethical considerations. Summarize what risks there are to this AI Product and what ethical considerations I should take into account.**"

Exercise

- ❑ Document risks
- ❑ Use AI to create a table of ethics questions
- ❑ Use AI to describe ways to mitigate the risks

10 Min Break

Financial Analysis

There are financial considerations to this AI Product.

Prompt: "**My AI Product is focused on <market> and I expect to make <amount of revenue> in my first 3 years. This AI Product will use scalable cloud services, is expected to reach 200,000 people each year with a 3% increase in reach every year. Write the financial model I need to ensure I can meet my market share and also make a profit. My AI Product will be subscription based and start at $4.99 / month**"

Exercise

- ❑ Identify financial model
- ❑ Use AI to determine Return on Investment (ROI)
- ❑ Use AI to determine a revenue forecast
- ❑ Use AI to identify break even time frame

NOTE: Use AI to make up numbers and forecasts. When starting a new business, you have to use markets and research. We don't have that. Let AI be creative.

Resource Management

Resources are a list of things you need to develop your product

- ❑ Use AI to identify which **cloud services** you need
- ❑ Use AI to identify how to **secure** your AI Product
- ❑ Use AI to identify which **marketing resources** you need
- ❑ Use AI to create a table of potential **start up costs** based on your business plan

What would your prompt be to create these things?

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Awards and Certificates

Term Definition

AI Product A tool or application powered by AI that performs tasks typically requiring human intelligence, such as natural language understanding or image recognition.

Cloud Services Online platforms providing scalable resources like storage and computing power, essential for running AI applications.

Confabulation (AI Hallucination) Instances where AI generates outputs that are plausible but factually incorrect or nonsensical.

DevSecOps An approach integrating development, security, and operations to deliver secure software more efficiently.

Ethical AI The practice of developing AI systems that align with ethical principles, considering societal impacts, fairness, and transparency.

Financial Model A plan detailing revenue streams, costs, ROI, and profitability for an AI product.

GANs (Generative Adversarial Networks) AI models composed of two neural networks—generator and discriminator—competing to create realistic data samples.

Generative AI A subset of artificial intelligence that uses algorithms like GANs and RNNs to generate new content such as text, images, or music, mimicking human creativity.

Innovation Framework A structured approach to designing and implementing AI solutions, focusing on ideation, product development, and business value creation.

Machine Learning (ML) A branch of AI focused on creating systems that can learn and improve from data without explicit programming.

Microservices A software architecture style where applications are built as collections of small, independent services that communicate over a network.

Natural Language Processing (NLP) AI technology that enables machines to understand, interpret, and generate human language.

Neural Networks Algorithms modeled after the human brain, designed to recognize patterns and solve complex problems through interconnected nodes.

Prompt Engineering The method of crafting inputs to maximize AI's effectiveness in generating accurate and useful outputs.

Resource Management Plan A strategic outline for allocating resources like cloud services, marketing tools, and security measures to support AI product development.

Risk Assessment The identification and analysis of potential risks in AI deployment, addressing technical, ethical, and operational concerns.

RNNs (Recurrent Neural Networks) A type of neural network designed for sequence prediction tasks, such as time-series data or language modeling.

Stern-Gerlach Experiment An experiment foundational in quantum mechanics, illustrating particle spin and measurement principles.

Validation & Regulation Processes ensuring that AI systems comply with legal and industry standards to guarantee safety, reliability, and ethical deployment.

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Prompt for Lithia

- i want this to use a different format. i teach 2 classes, Value of AI in Business, and AI and Agent ready data. i have 4 phases, business case and ideation, production development, data and ai, presentation of the final app. the first class asks the students to create a business case, the type of jobs that will be necessary, their descriptions, a 30 second marketing pitch and a list of financials they expect to achieve. In doing this, they create a marketing pitch. I want this app here, to walk them through it and based on a description of what they want to accomplish I want you the ai to present them some options for products. The product development phase will consist of developing documentation and createing concepts with mock ups. The students will need to know how to get data. The data phase will be about collecting datasets that are either created or real and then you as an ai can use them in the app. At the same time the students will need to train an AI and enhance data with metadata to help protect and classify it. Finally the presentation phase, each student will put their app together with the data and quality factors and present it.

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